Summary in Graph

Exam Summary (GO Classes CS Test Series | C-Programming | Topic Wise Test 3)

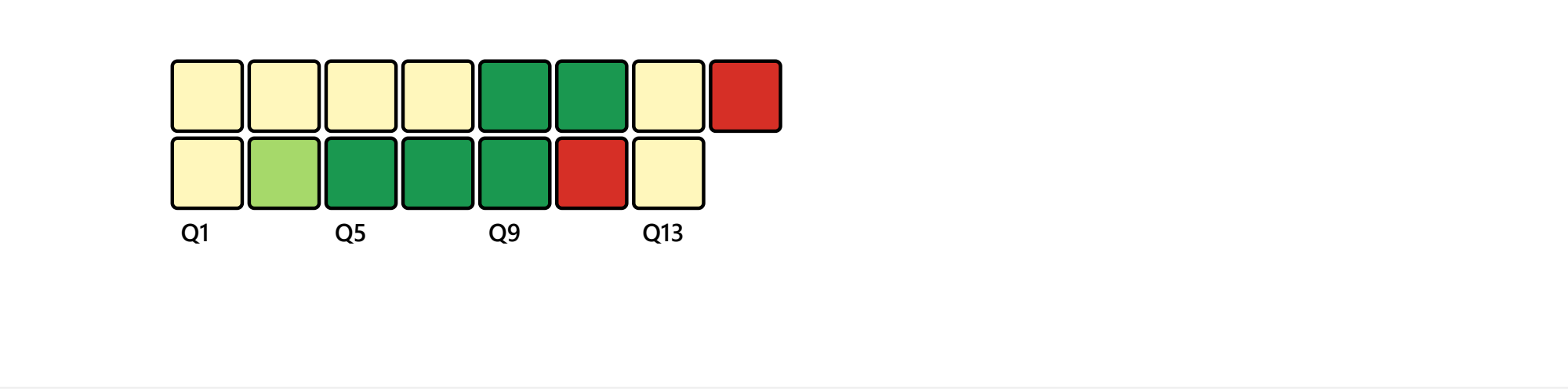
38.7%
9.67/25

6 Correct
6 Wrong
3 Skipped
44m of 45m

Qs. Attempted:	12	4 + 8
Correct Attempts:	6	1 + 5
Incorrect Attempts:	6	3 + 3
Correct Marks:	11	1 + 10
Penalty Marks:	1.33	0 + 1.33
Resultant Marks:	9.67	1 + 8.67
Total Questions:	15	5 + 10
Total Marks:	25	5 + 20
Exam Duration:	45 Minutes	
Time Taken:	44 Minutes	

Individual Question Statistics

View Order: Sorted Order
Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (100) out of 1001 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C | Medium (45%)

In which of the following case(s) character array must end with null char?

A. `char c[] = "abcd";`
B. `char c[] = {'a', 'b', 'c', 'd'};`
C. `char c[5] = "abcd";`
D. `char c[4] = "abcd";`

Your Answer: B;D | Correct Answer: A;C | Incorrect | Time taken: 00min 59sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C | Hard (22%)

What will be the output of a given program in a little endian system? (Assume int takes 4 bytes)

```c
#include
int main()
{
    char t[] = "abcdefg";
    int *p = t;
    p++;
    char *c = p;
    printf("%c", *c);
}
```

A. b
B. d
C. e
D. f

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 01min 53sec | Discuss

Q #3 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C | Hard (12%)

Which of the following may lead to a runtime error for the given declaration of p?

```c
int (*p)[5];
```

A. `p[1][1]`
B. `*p[1]`
C. `(*p)[1]`
D. `p[1]`

Your Answer: A | Correct Answer: A;B | Incorrect | Time taken: 02min 02sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C | Hard (26%)

Consider the following C program.

```c
#include<stdio.h>
void fun (int a[5]){
    for (int i =0; i<8; i++)
        a[i] = a[i]+1;
}
main ()
{
    int a[8] = {1,2,3,4,5,6,7,8};
    fun(a);
    for (int i =0; i<8; i++)
    printf("%d", a[i]);
}
```

Which of the following is/are true about the given program?

A. Program will exhibit a compiler error since array “a” is of size 5 in function fun() but for loop in fun() runs till 8.
B. Program will exhibit run time error since array “a” is of size 5 in function fun() but for loop in fun() runs till 8.
C. On running the program, the output will be 23456789
D. None of the above

Your Answer: C | Correct Answer: C | Correct | Time taken: 02min 37sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C | Hard (15%)

Which of the following is/are valid assignment(s) of the pointer for the given declaration. An assignment is valid if both pointers are of the same type (pointers are compatible) and get successfully compiled with no warning or error.

```c
int a[4][4];
int *t, **pp;
```

A. `pp = a;`
B. `pp = &a;`
C. `pp = *a;`
D. `t = a[2];`

Your Answer: A;B;D | Correct Answer: D | Incorrect | Time taken: 01min 30sec | Discuss

Q #6 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Hard (18%)

Consider the given declaration

```c
int arr[3][4];
int (*p)[12];
p = malloc(sizeof(int) * 24); // assume malloc() is successful
```

```c
sizeof(int) = 4 bytes and sizeof(int *) = 8 bytes.
```

What will be the output of the following lines of C code ?

```c
printf("%d " , sizeof(arr[2]));
printf("%d ", sizeof(p));
printf("%d", sizeof(*p));
```

A. $8\ 8\ 8$
B. $8\ 8\ 48$
C. $16\ 8\ 96$
D. $16\ 8\ 48$

Your Answer: D | Correct Answer: D | Correct | Time taken: 05min 12sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C | Hard (26%)

The C code printed below is a version of the C string library call strlen(), which returns the length of a given string.

```c
unsigned int mystrlen(char *c)
{
    unsigned int i =0;
    /* Your code here. */
    return i;
}
```

Which line of code creates a working and accurate version of strlen()?

A.
```c
while(*c!='\0'){
    i++;
    c++;
  }
```
B. `while(*c++!='\0') i++;`
C. `while(*c++ && i++);`
D. `while (c[i++] != '\0');`

Your Answer: A;B;D | Correct Answer: A;B | Incorrect | Time taken: 04min 37sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (34%)

What will be the output of the following lines of C code?

```c
int a[] = {1,2,3,4};
int *p = (int *)(&a+1);
printf("%d", *(p-1));
```

A. 1
B. 2
C. 3
D. 4

Your Answer: D | Correct Answer: D | Correct | Time taken: 01min 24sec | Discuss

Q #9 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Easy (68%)

What will be the output of the following C code?

```c
#include<stdio.h>
void strXpos(char * dest, char * src) {
    if (src[0] == '\0') {
        dest[0] = '\0';
        return;
    }
    if (src[1] == '\0') {
        dest[0] = src[0];
        dest[1] = src[1];
        return;
    }
    dest[0] = src[1];
    dest[1] = src[0];
    strXpos(dest + 2, src + 2);
}
int main() {
    char s1[10] = "GOGATE";
    char s2[10] = "GO";
    strXpos(s2, s1);
    printf("%s", s2);
}
```

A. GOGATE
B. OGAG
C. OGAGET
D. GOAGET

Your Answer: C | Correct Answer: C | Correct | Time taken: 04min 05sec | Discuss

Q #10 Numerical Type | Award: 2 | Penalty: 0 | Programming in C | Hard (28%)

What is the maximum value of i that will not cause this code to overflow?

It is given that $\text{sizeof(int)} = 4$ and $\text{sizeof(int *)} = 8$.

```c
int *p = malloc(sizeof(int *) * 2);
p[i] = 2501;
```

Your Answer: 3 | Correct Answer: 3 | Correct | Time taken: 02min 26sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (58%)

What will be the output of the following program?

```c
#include<stdio.h>
#include<string.h>

struct _info {
    char **strs;
};
typedef struct _info info;
main(){
    info arr_str[2];
    char *strs1[] = {"GO", "Classes"};
    char *strs2[] = {"GATE", "2023"};
    arr_str[0].strs =strs1;
    arr_str[1].strs =strs2;
    info **da, *a;
    a =arr_str;
    da = &a;
    for (int i = 0; i<2; i++)
        for (int j = 0; j< 2; j++){
            printf("%s ", (((*da)+i)->strs)[j]);
        }
}
```

A. GO GATE Classes 2023
B. GO Classes GATE 2023
C. GATE GO 2023 Classes
D. GATE 2023 GO Classes

Your Answer: B | Correct Answer: B | Correct | Time taken: 03min 44sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (42%)

What will be the output of the program below-

```c
int u[] = {1, 7, 8, 3};
int v[] = {-4, 9, 33};
int w[] = {12, 2, 57};

struct t{
    char *c;
    int *t;
}go[] = {"GATE", u, "GOClasses", v, "GATEOVERFLOW", w};

main()
{
    struct t *p = go;
    printf("%s ", ++p -> c);
    printf("%d ", ++*p -> t);
    printf("%d ", *++p[0].t);
    printf("%s ", (++p) -> c);
}
```

A. ATE 1 9 GATEOVERFLOW
B. GATE -4 2 GATEOVERFLOW
C. ATE 2 7 GOClasses
D. None of the above

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 06min 27sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Hard (22%)

Assume that variables $p1$, $p2$, and n are defined as follows:

```c
int *p1 = (int *) malloc(2 * sizeof(int));
int *p2 = p1 + 1;
int n = 1;
```

Given 4 programs and 4 types of issues below, check which program is likely to cause which type of issue.

Prog1 :	`free(p1);`
	`*(p2 - 1) = n;`
	`free(p2);`
		A. Using unallocated memory

Prog2 :	`free(p1);` `p2 -= 1;`	B. Memory leak
Prog3 :	`*(p2 + 1) = n;` `free(p1);`	C. Dangling Pointer

Prog4 :	`p1 = \&n;`
	`*p1 = n;`
	`free(p1);`
		D. No Issue

A. Prog1-C, Prog2-C, Prog3-A, Prog4-D
B. Prog1-C, Prog2-D, Prog3-A, Prog4-B
C. Prog1-C, Prog2-D, Prog3-A, Prog4-D
D. Prog1-D, Prog2-C, Prog3-A, Prog4-B

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 04min 13sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (34%)

Consider struct alignment rules as described below –

1. Each primitive data (int, char, float etc) type requiring K bytes must be stored at the address must be multiple of K.
2. Address of structure and structure length must be multiples of $\mathrm{K_{max}}$. Where $\mathrm{K_{max}}$ is the largest size of any primitive element of the struct.

Also, we define internal and external fragmentation of struct as follows–

- Internal fragmentation is padding used between any two primitive data types of the struct.
- External fragmentation is padding used at the end of the last element of struct to fulfill 2^{nd} rule requirement of alignment above.

For a given struct below, what will be internal and external fragmentation, respectively? Consider that the system has 8 bytes of address lines, 4 bytes of integers, 2 bytes of short, and 1 byte of the character.

```c
typedef struct {
    int num_donors;
    char** donor_names;
    char has_soft_seats;
    char location_name[10];
    short height;
} cse_building;
```

A. $4, 3$
B. $5, 2$
C. $0, 7$
D. $4, 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 01min 08sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C | Medium (32%)

Consider the following declarations of variables in a system having 4 bytes for integers.

```c
int a[4][4];
int (*p1)[2];
int (*p2)[2];
int (*t1)[4];
int (*t2)[4];
```

Suppose array a has base address of 1000 and initial values of $p1, p2, t1,$ and $t2$ are $1040, 1064, 1044$ and 1076 respectively.

What will be the value of $p2 - p1$ and $t2 - t1$ respectively?

A. $6, 8$
B. $3, 2$
C. $12, 8$
D. $32, 24$

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 02min 05sec | Discuss

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